[METRIC]
A-A-52609
May 27, 1996
SUPERSEDING
MIL-P-62680(AT)
9 October 1992

COMMERCIAL ITEM DESCRIPTION

POWDER, NICKEL-CHROMIUM/ALUMINUM COMPOSITE

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This CID covers nickel-chromium/aluminum composite powder used in plasma applications. This powder is primarily intended for use as a plasma spray deposited material providing increased oxidation and hot corrosion resistance.

2. SALIENT CHARACTERISTICS

2.1 Materials. Unless otherwise specified herein, the materials used shall be in accordance with the manufacturer's materials specification. The use of recovered materials made in compliance with regulatory requirements is acceptable providing all requirements of this CID are met (see 3.1).

2.1.1 Particle size. The particle size of the powder shall be as specified in table I.

2.1.2 Chemical composition. The chemical composition of the material shall conform to the percent by weight for each element as specified in table II.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.

AMSC N/A FSC 9630

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I . Particle size distribution by weight.

Sieve designation 1/		Percentage (%) by weight (Maximum)	Weight passed through sieve	Weight retained on sieve
Micron	Mesh			
150	100	1	-	X
125	120	10	-	X
45	325	20	X	-

1/ Sieve designation shall be in accordance with ASTM E11.

TABLE II. Chemical composition.

Elements	(%) by weight	
	Minimum	Maximum
Manganese	-	3.0
Chromium	15.5	21.5
Aluminum	4.5	7.5
Silicon	-	2.0
Iron	-	1.5
Carbon (after removing organic solids)	-	0.3
Organic solids, if determined	-	3.5
Nickel	69.5	-

2.1.3 Properties. The powder shall meet the following properties when applied (plasma sprayed) to iron and iron base alloys.

a. Bond strength. The bond strength of the plasma-sprayed material shall be not less than 36.5 megapascals (mPa) [5300 pounds per square inch (psi)].
b. Cup test. The amount of draw before spalling of the coating shall not be less than 4.5 millimeters (mm) [0.175 inch (in.)].

2.2 Unit containers. Unless otherwise specified (see 6.2), the unit containers shall be capable of holding the following quantities of powder: 0.5, 1.0, 2.0, 5.0 and 10.0 kilograms(kg) [1, 2.5, 5, 12.5 and 25 pounds (lbs)]. The containers shall be sealed moisture-proof units and large enough to permit mixing the powder within before use. The containers shall be capable of protecting the powder against damage or contamination until it is ready to be used.

2.3 Identification markings. The identification markings shall be permanent and legible and shall include, as a minimum, the manufacturer's CAGE code, part number, lot number and the part or identification number (PIN) (see 6.3).

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2.3.1 Labeling. Unit containers and packages shall be conspicuously labeled " CAUTION: Containers contain powder aluminum. Use approved respirator".

2.3.2 Container. Each container shall be marked "Nickel-Chromium, Aluminum Powder".

3. REGULATORY REQUIREMENTS

3.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The offeror/contractor is responsible for the performance of all inspections (examinations and tests).

4.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Addresses for obtaining copies of referenced documents.

6.1.1 Government documents. Copies of the Code of Federal Regulations (CFR) are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

6.1.2 Non-Government documents. Copies of ASTM E11, "Standard Specification for Wire Cloth and Sieves for Testing Purposes" (DOD Adopted) are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

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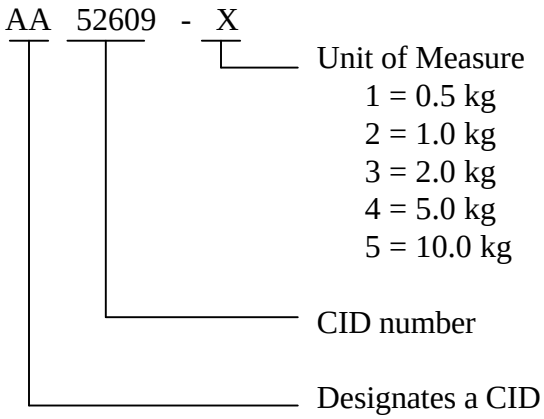
6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Issue of Department of Defense Index of Specifications and Standards (DoDISS) to be cited in the solicitation, and if required, the specific issue of individual documents referenced.
c. Unit container and PIN number.
d. Unit container size if other than specified.
e. Packaging requirements.

6.3 Part or identification number (PIN). The PINs to be used for powder acquired to this CID are created as follows:



6.4 Cross reference data. Power conforming to this CID is interchangeable/substitutable with powder conforming to MIL-P-62680(AT). Cross reference to former part number is indicated in table III.

TABLE III. Cross reference.

Unit container size	PIN	Former PIN no.
0.5 kg	AA52609-1	M62680-0.5 kg
1.0 kg	AA52609-2	M62680-1.0 kg
2.0 kg	AA52609-3	M62680-2.0 kg
5.0 kg	AA52609-4	M62680-5.0 kg
10.0 kg	AA52609-5	M62680-10.0 kg

6.5 Supersession data. This CID supersedes Textron Lycoming (Allied Signal) internal specification M3956E, dated 21 October 1986.

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MILITARY INTERESTS:

Custodian:
Army - AT

CIVIL AGENCY COORDINATION ACTIVITY:
GSA-FSS

Preparing Activity:
Army- AT

(Project 9630- 0038)

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